UNIT 6: DIVERSITY OF LIFE FORMS
6.6 – INVERTEBRATES

Biology 9 Mr. Queenan
Ch. 24.1-24.2 (pg. 800-809)

Invertebrates Outline

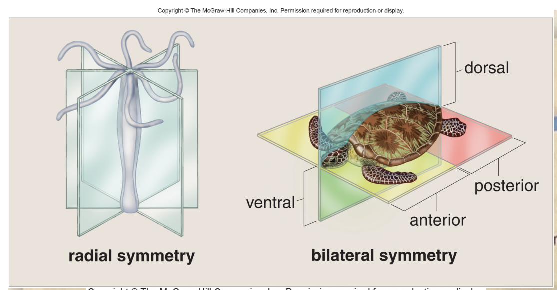
- Characteristics of Animals
  - Evolution
  - Symmetry of Body Plans
  - Development
- Invertebrate Diversity
  - Simple Invertebrates
  - Radiata
  - Lophotrochozoans
  - Ecdysozoans
  - Deuterostomes

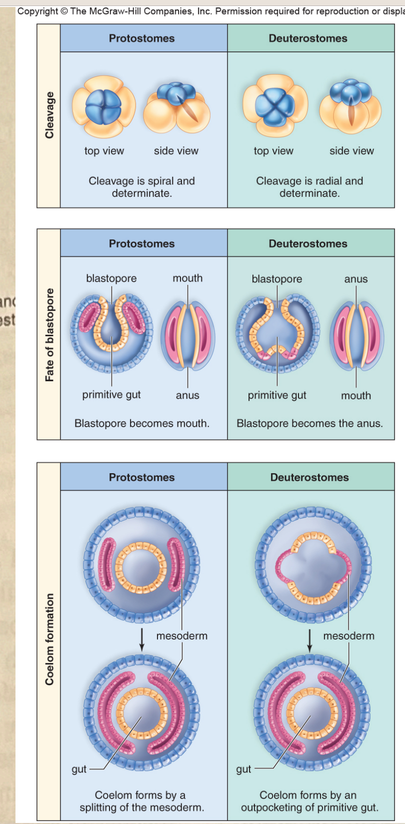
Characteristics of Animals

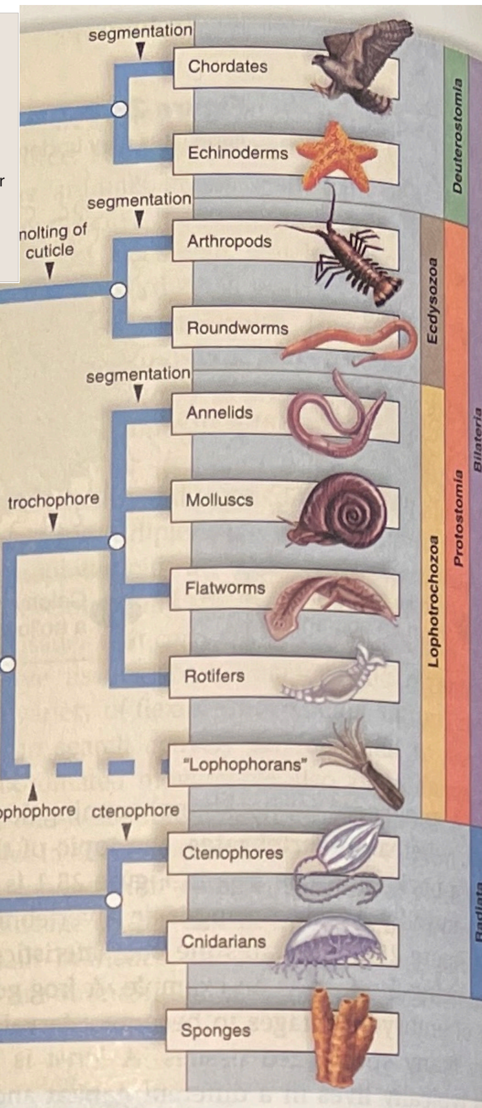
- All animals have a number of common features
  - Heterotrophs, that must ingest food, and digest internally
  - Cells lack cell walls
  - Generally mobile
  - Coordinated movement using nerves and muscles
  - Reproduce sexually
    - Invertebrates
      - Do not have a backbone
    - Vertebrates
      - Have a spinal cord or backbone at some stage of their life running down the center of their back
  - Most likely descended from an ancestral flagellate colony, where cells took specialized roles
  - Wide diversity in body plans, but most members have varying developmental plans controlled by *Hox* gene family

Animal Phylogenetic Tree

- Major branchings in the tree occur based on the following criteria:
  - Symmetry
    - Can be **radial** (like a wheel) or **bilateral** (left and right along a single longitudinal line)
  - Embryonic Development
    - Organisms can have tissue layers
      - **Diploblastic** - 2 layers
        - Tissues developed bu no specialized organs
      - **Triploblastic** – 3 layers
        - Ecto, meso, & endoderm
        - Organisms either **protostomes** (mouth develops before anus) or **deuterostomes** (anus develops before mouth)
    - Other branchings between protostomes and deuterostomes based on cleavage of cells, blastula formation, and coelom development

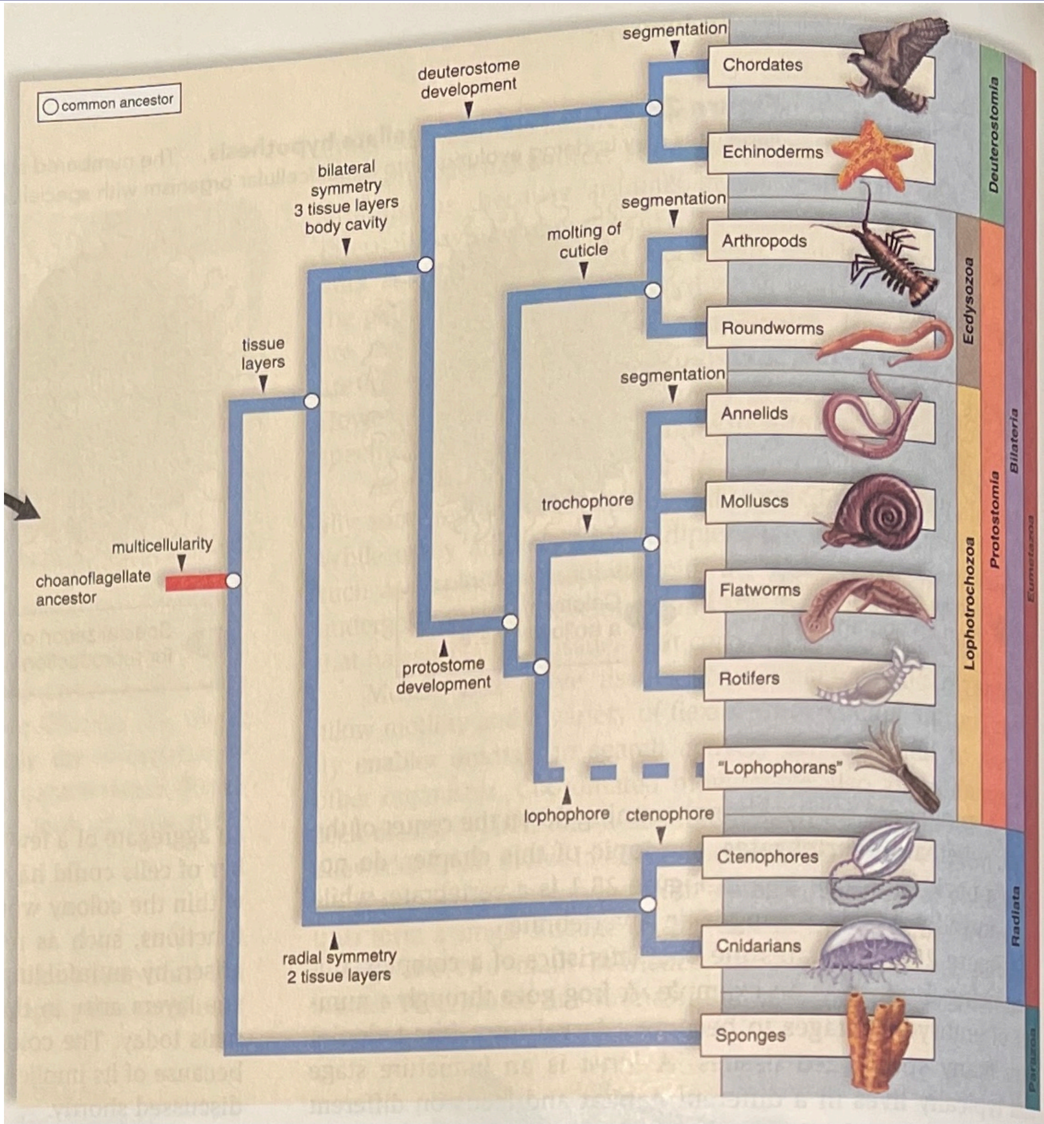






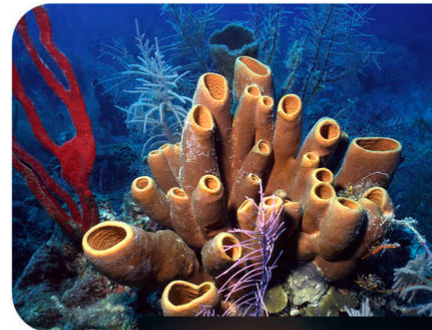
Diversity of Invertebrates

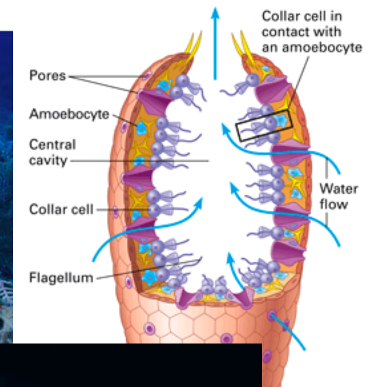
- Simple Invertebrates
- Radiata
- Protostomia
  - Lophotrochozoa
  - Ecdysozoa
- Deuterostomia



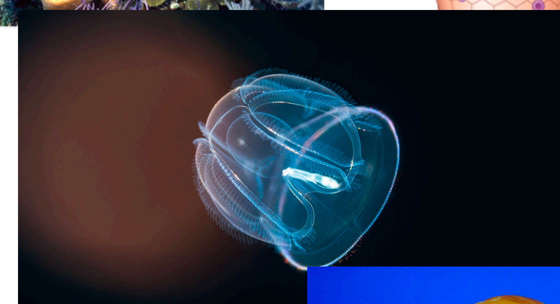
Diversity – Simple Invertebrates & Radiata

- **Sponges**
  - Only animal to lack true tissue
  - Body perforated by pores
  - Interior canal contains flagellated cells
  - Filter feeder
  - Spicule structure used for identification
  - Reproduce sexually and asexually

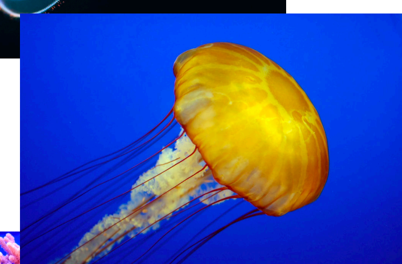




- **Comb Jellies**
  - Solitary, free swimming marine animal
  - Use beating cilia to move
    - A few cm to 1.5 m in length
  - Body made of transparent jellylike substance – **mesoglea**
  - Most do not have stinging properties; adhere to prey
  - Some bioluminescent

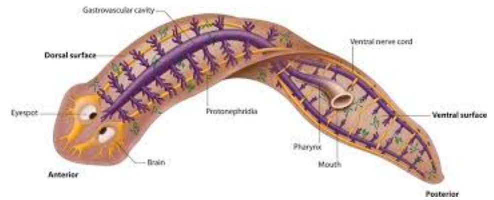


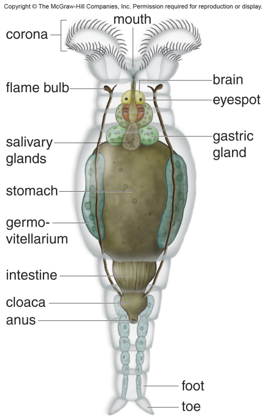
- **Cnidarians**
  - Most have stinging cells **– cnidocytes**
  - 2 layered sac (internal is digestive cavity)
    - Also serves as hydrostatic skeleton
  - Body form **polyp** -cup up; **medusa** - cup down
  - Sea anemone, coral, hydrozoans, jellyfish, hydra



Diversity - Lophotrochozoans

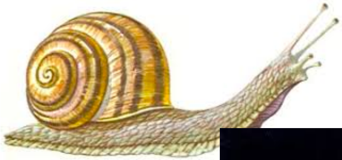
- Bilaterally symmetrical at some point in their life
- Divided into Lophophorans and Trochozoans
  - Lophophorans
    - Aquatic, with mouth surrounded by ciliated tentacle-like structure – **lophophore**
    - Bryozoans, Brachiopods, & Phoronoids
  - Trochozoans
    - Flatworms (Platyhelminthes)
      - Free living – planarian
        - Well developed excretory system; sexual reproducing (hermaphroditic); ladder type nervous system
      - Parasitic - Flukes, tapeworms
        - Live off host system for survival
    - Rotifers (Rotifera)
      - Have a crown of cilia on their heads.





Diversity - Lophotrochozoans

- Trochozoans - continued
  - Molluscs (Mollusca)
    - 3 part body plan. More developed body plan
    - Bivalve (Bivalvia)
      - 2 part shell
      - Clams, scallops, mussels
    - Gastropods (Stomach foot)
      - Long, flattened foot and coiled shell
      - Snails, slugs, conch, nudibranch
    - Cephalopods (Head Foot)
      - Well developed brains
      - Squid, octopus
  - Annelids (Annelida)
    - Segmented worms
      - Repeating body segment pattern
      - Closed circulatory system
      - Earthworms, marine worms, leeches

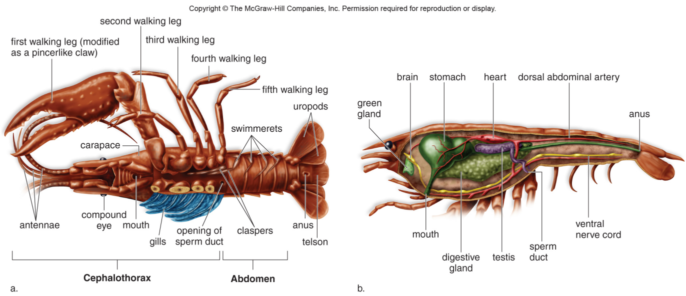


Diversity - Ecdysozoans

- Large group containing roundworms and arthropods
  - Roundworms (Nematoda)
    - Non-segmented nematodes
    - Can be free-living or parasitic
    - *Caenorhabditis elegans* is a model research organism
    - *Ascaris* infections can occur in humans and other animals
    - *Trichinosis* serious infection from eating uncooked pork
    - Filarial worms and pinworms also common parasitic roundworms

Diversity - Ecdysozoans

- Arthropods (Arthropoda)
  - Members have exoskeleton, jointed appendages, brain and nervous system, and a variety of respiratory organs
    - Some go through a process of metamorphosis
  - Phylum contains:
    - Crustaceans – hard crusty exoskeleton made of chitin and calcium carbonate. Head with compound eyes, and 5 pairs of appendages
      - Copepod, krill, crayfish
    - Millipedes & Centipedes – Distinguished by # of pairs of leg on body segments
    - Insects – have head, thorax, and abdomen. Head has sense organs, thorax has 3 pairs of legs and 1 or 2 pairs of wings
      - Grasshopper, bee, butterfly
    - Chelicerates – First pair of appendages pincher-like for feeding and defense, second pair can have multiple functions. Have a cephalothorax (fused head and thorax) followed by abdomen.
      - Horseshoe crab, Arachnids (ticks, mites, spiders, scorpions)



Diversity - Deuterostomes

- Include Echinoderms and Chordates. All echinoderm are invertebrates. Most chordates are vertebrates
- Echinoderms
  - Primarily bottom dwelling marine animals.
  - Range in size from 1cm to 2m
  - 5-pointed radial symmetry as adults (bilateral as larva)
  - Endoskeleton of spiny, calcium-rich plates called ossicles
  - Water vasuclar system for locomotion, feeding, gas exchange, and sensory reception
    - Include sea stars, sea cucumbers, sea urchins and sand dollars
- The chordate invertebrates will be discussed in the next lesson

Overview

Clearer version of the invertebrate overview can be found on p526

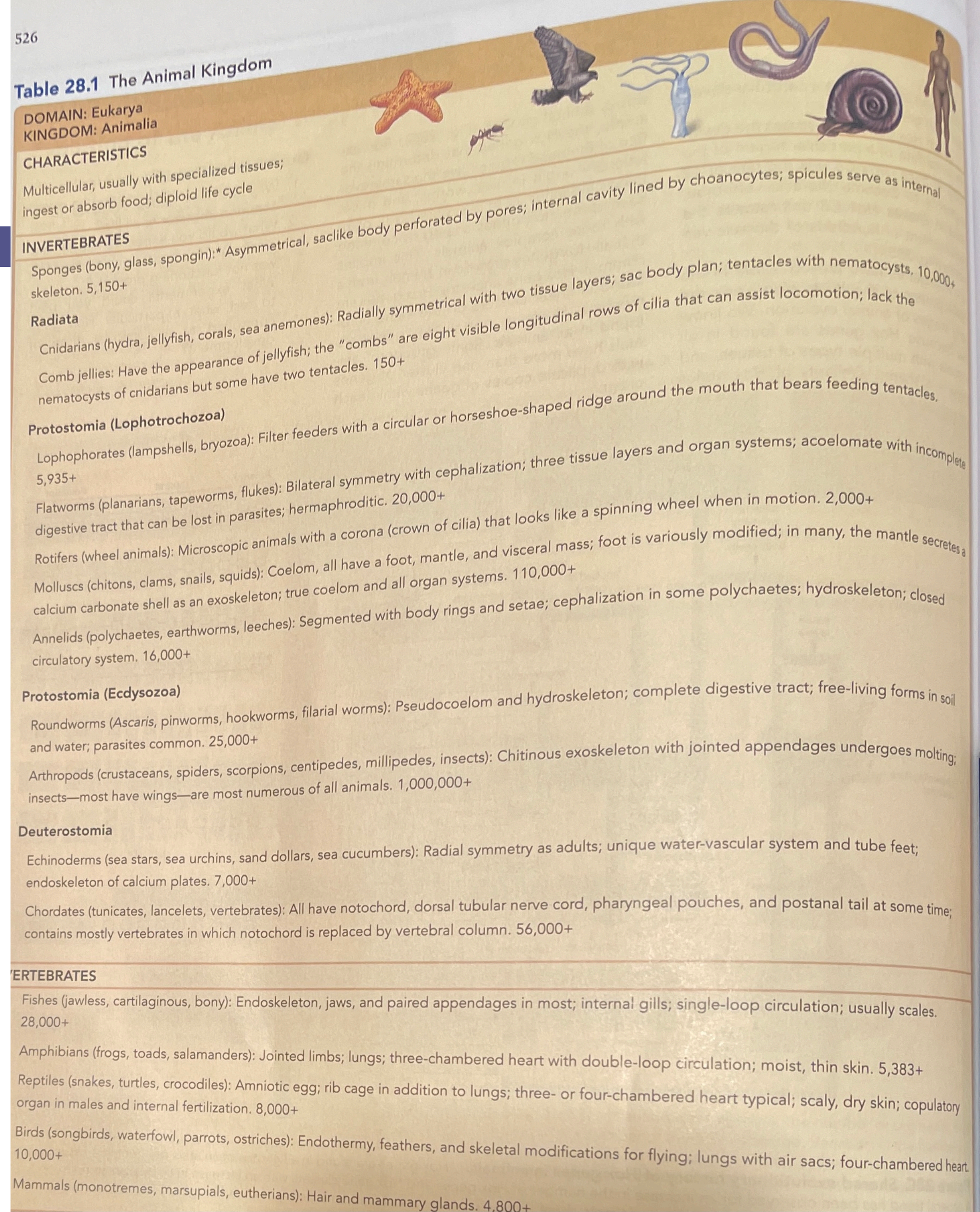


Table 28.1 The Animal Kingdom

DOMAIN: Eukarya
KINGDOM: Animalia

CHARACTERISTICS

Multicellular, usually with specialized tissues; ingest or absorb food; diploid life cycle

INVERTEBRATES

Sponges (bony, glass, spongin):* Asymmetrical, saclike body perforated by pores; internal cavity lined by choanocytes; spicules serve as internal skeleton. 5,150+

Radiata

Cnidarians (hydra, jellyfish, corals, sea anemones): Radially symmetrical with two tissue layers; sac body plan; tentacles with nematocysts. 10,000+

Comb jellies: Have the appearance of jellyfish; the "combs" are eight visible longitudinal rows of cilia that can assist locomotion; lack the nematocysts of cnidarians but some have two tentacles. 150+

Protostomia (Lophotrochozoa)

Lophophorates (lampshells, bryozoa): Filter feeders with a circular or horseshoe-shaped ridge around the mouth that bears feeding tentacles. 5,935+

Flatworms (planarians, tapeworms, flukes): Bilateral symmetry with cephalization; three tissue layers and organ systems; acoelomate with incomplete digestive tract that can be lost in parasites; hermaphroditic. 20,000+

Rotifers (wheel animals): Microscopic animals with a corona (crown of cilia) that looks like a spinning wheel when in motion. 2,000+

Molluscs (chitons, clams, snails, squids): Coelom, all have a foot, mantle, and visceral mass; foot is variously modified; in many, the mantle secretes a calcium carbonate shell as an exoskeleton; true coelom and all organ systems. 110,000+

Annelids (polychaetes, earthworms, leeches): Segmented with body rings and setae; cephalization in some polychaetes; hydroskeleton; closed circulatory system. 16,000+

Protostomia (Ecdysozoa)

Roundworms (*Ascaris*, pinworms, hookworms, filarial worms): Pseudocoelom and hydroskeleton; complete digestive tract; free-living forms in soil and water; parasites common. 25,000+

Arthropods (crustaceans, spiders, scorpions, centipedes, millipedes, insects): Chitinous exoskeleton with jointed appendages undergoes molting; insects—most have wings—are most numerous of all animals. 1,000,000+

Deuterostomia

Echinoderms (sea stars, sea urchins, sand dollars, sea cucumbers): Radial symmetry as adults; unique water-vascular system and tube feet; endoskeleton of calcium plates. 7,000+

Chordates (tunicates, lancelets, vertebrates): All have notochord, dorsal tubular nerve cord, pharyngeal pouches, and postanal tail at some time; contains mostly vertebrates in which notochord is replaced by vertebral column. 56,000+

ERTEBRATES

Fishes (jawless, cartilaginous, bony): Endoskeleton, jaws, and paired appendages in most; internal gills; single-loop circulation; usually scales. 28,000+

Amphibians (frogs, toads, salamanders): Jointed limbs; lungs; three-chambered heart with double-loop circulation; moist, thin skin. 5,383+

Reptiles (snakes, turtles, crocodiles): Amniotic egg; rib cage in addition to lungs; three- or four-chambered heart typical; scaly, dry skin; copulatory organ in males and internal fertilization. 8,000+

Birds (songbirds, waterfowl, parrots, ostriches): Endothermy, feathers, and skeletal modifications for flying; lungs with air sacs; four-chambered heart. 10,000+

Mammals (monotremes, marsupials, eutherians): Hair and mammary glands. 4,800+